Exp. No: 5

Electric Potential Difference between Two Points in Free Space

Aim:

To write program for Electric potential difference between two points in free space medium using MATLAB.

Objectives:

1. To determine the electric potential difference between two points separated by a distance R in free

space.

2. To plot the variation of potential difference with separation distance.

3. To relate the potential gradient to electric field intensity for validation.

Test case (i):

Determination of electric potential difference between two points for a given charge

Program:

```
clc;

clear;

q=input("Enter the q value:");

epsilon=8.854*10^-12;

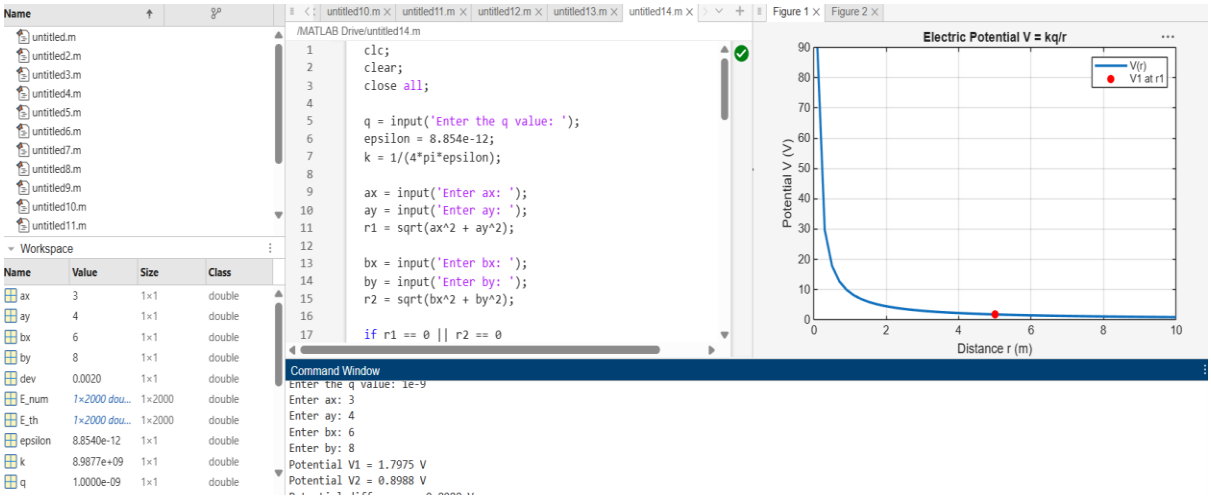
k=1/(4*pi*epsilon);

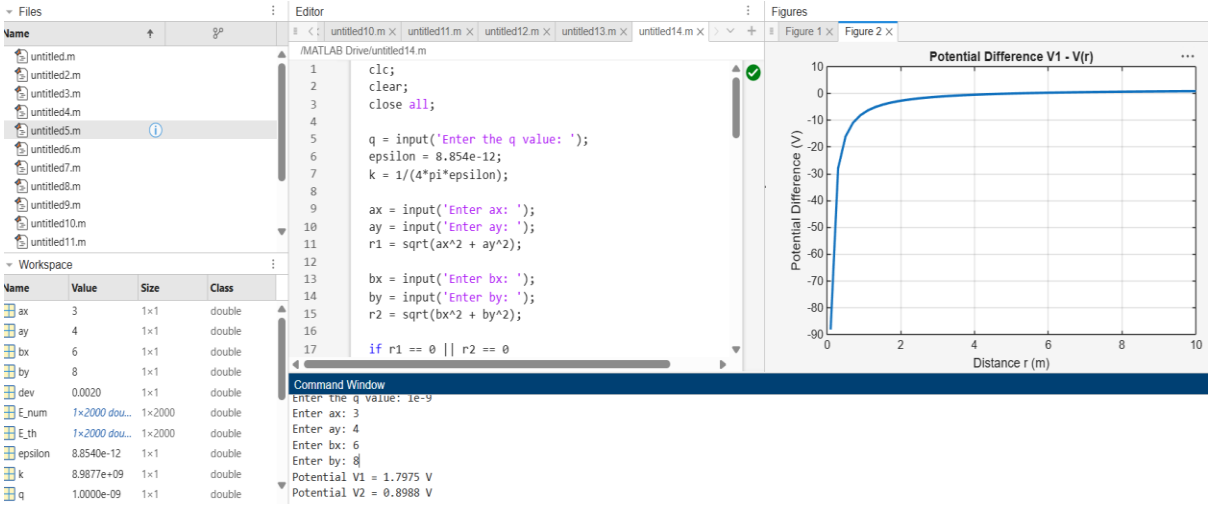
ax=input('Enter the value of ax=');
```

```
ay=input('Enter the value of ay=');
r1=sqrt(ax^2+ay^2);
v1=k*q/r1;
disp('The value of v1');
disp(v1);
bx=input ('Enter the value of bx=');
by=input ('Enter the value of by=');
r2=sqrt(bx^2+by^2);
v2=k*q/r2;
disp('The value of v2');
disp(v2);
v=v1-v2;
disp ('The electric potential difference is');
disp (v);
% Plot: potential versus radial distance and potential difference between r1 and r
r = linspace(0.1,10,50);
V = k*q./r;
% V = kq/r
figure;
plot(r,V,'LineWidth',2); hold on;
plot(r1,v1,'ro','MarkerFaceColor','r'); grid on;
xlabel('Radial distance r (m)'); ylabel('Potential V (V)');
title('Potential V = kq/r'); legend('V(r)','V1 at r1');
figure;
```

```
plot(r,k*q*(1/r1 - 1./r),'LineWidth',2); grid on;
xlabel('Radial distance r (m)'); ylabel('Potential difference V1 - V(r) (V)');
title('Potential difference between r1 and r');
% Independent check: E = -dV/dr (numerical gradient) compared with E = kq/r^2
rc = linspace(1,10,2000);
E_num = -gradient(k*q./rc,rc);
E_th = k*q./rc.^2;
dev = max(abs(E_num(2:end-1) - E_th(2:end-1))./E_th(2:end-1))*100;
fprintf('Maximum deviation between -dV/dr and kq/r^2 = %.4f %%\n', dev);
```

output:





Test case (ii):

A positive charge with fixed r_1 = 1 m and r_2 varying.

Program:

```
clc;

epsilon=8.854e-12;

k=1/(4*pi*epsilon);

q=1e-6;

r1=1;

r2=linspace(1.1,10,100);

V=k*q*(1/r1 - 1./r2);

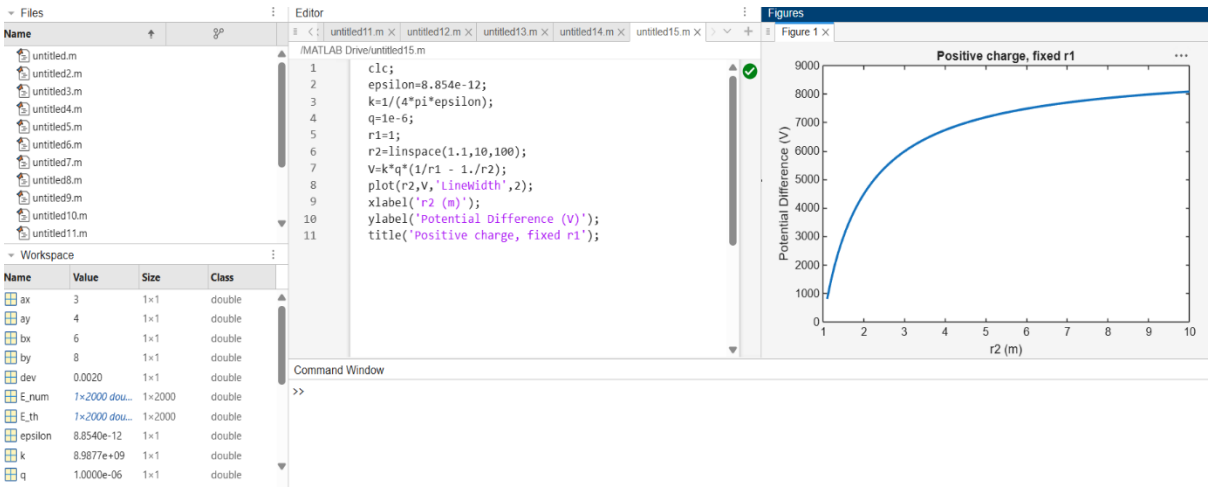
plot(r2,V,'LineWidth',2);

xlabel('r2 (m)');

ylabel('Potential Difference (V)');

title('Positive charge, fixed r1');
```

output:



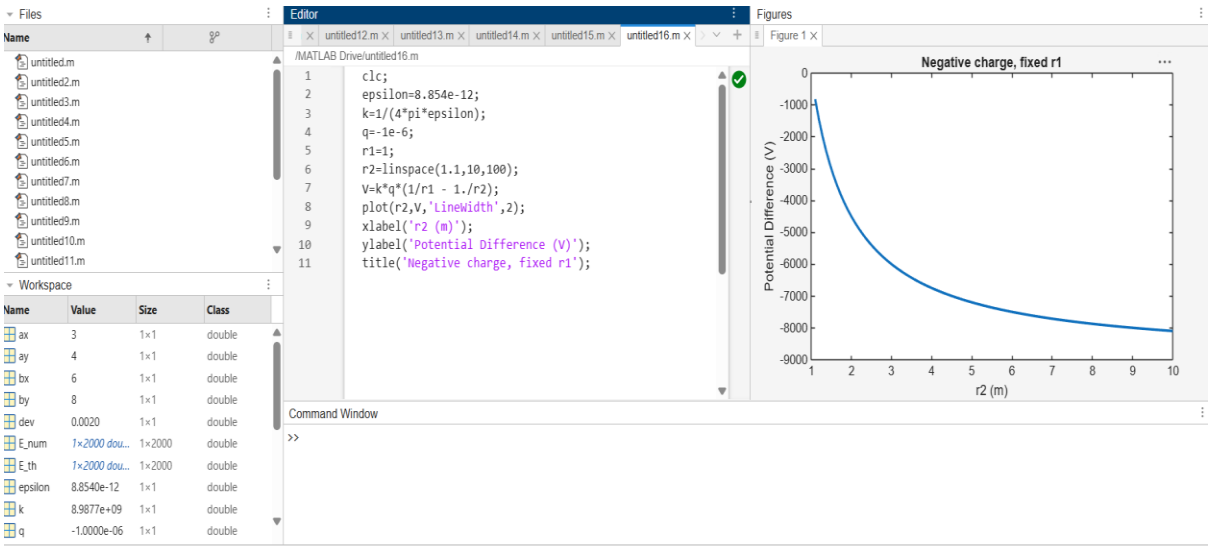
Test case (iii):

A negative charge with fixed r_1 and varying r_2:

Program:

```
clc;
epsilon=8.854e-12;
k=1/(4*pi*epsilon);
q=-1e-6;
r1=1;
r2=linspace(1.1,10,100);
V=k*q*(1/r1 - 1./r2);
plot(r2,V,'LineWidth',2);
xlabel('r2 (m)');
ylabel('Potential Difference (V)');
title('Negative charge, fixed r1');
```

output:



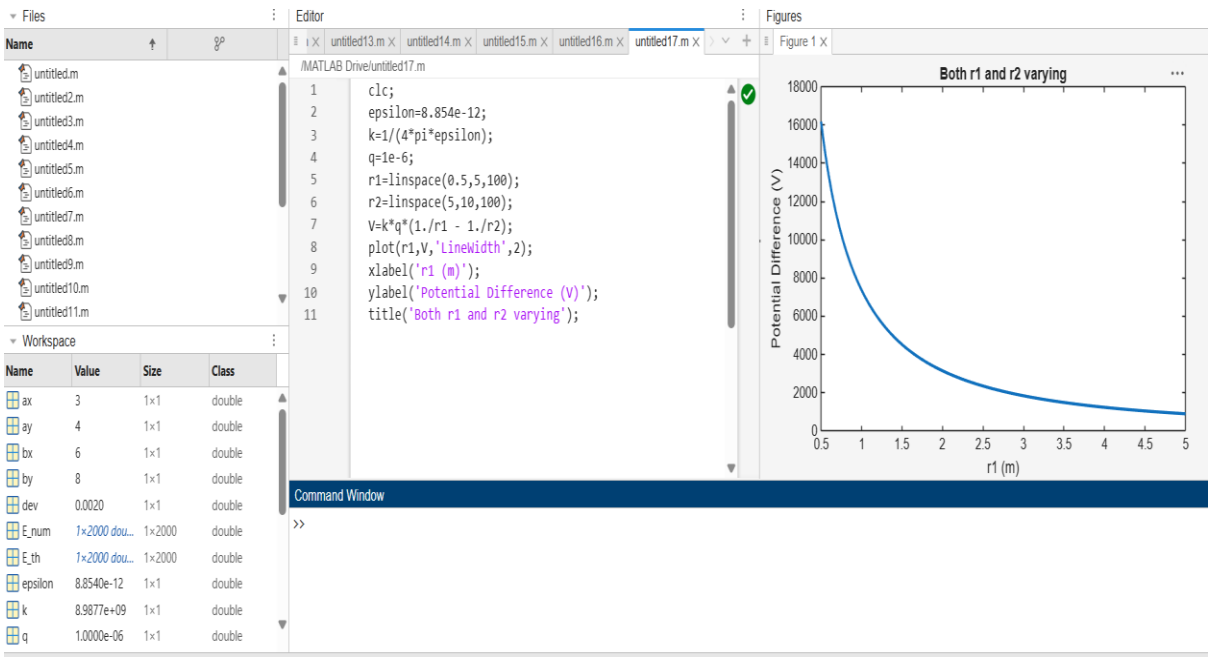
Test case (iv) :

Both r_1 and r_2 varying with positive charge 3:

Program:

```
clc;
epsilon=8.854e-12;
k=1/(4*pi*epsilon);
q=1e-6;
r1=linspace(0.5,5,100);
r2=linspace(5,10,100);
V=k*q*(1./r1 - 1./r2);
plot(r1,V,'LineWidth',2);
xlabel('r1 (m)');
ylabel('Potential Difference (V)');
title('Both r1 and r2 varying');
```

output:



Rubrics:

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	**Define charge, reference and observation points**	Define charge, reference and observation points	Correctly identifies charge distribution, point locations, distance and reference convention.	Setup is correct with minor clarification needed.	Some input or reference details are unclear.	Uses an incorrect setup or reference.	25	2.5	
2	**Calculate potential and potential difference**	Calculate potential and potential difference	Calculates potential at both points and signed difference correctly with units.	Results are mostly correct with minor arithmetic errors.	Partial calculations require support.	Potential calculations are incorrect or absent.	25	2.5	
3	**Verify potential relationship**	Verify potential relationship	Checks results against analytical equations, and, where relevant, the relation between E and potential.	Validation is mostly correct with minor omissions.	Validation is incomplete.	No valid theoretical check is provided.	30	3	
4	**Interpret and report findings**	Interpret and report findings	Explains the sign and physical meaning of potential difference with clear tables/conclusion.	Explanation is correct with minor omissions.	Explanation is limited or unclear.	Cannot explain or document the result.	20	2	
	Total						**100**	**10**	